CentraleSupélec

Relaxation

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1 Introduction

In the context of fast-growing AI-driven frameworks coupled with increasing compute resources, bigger and bigger deep learning networks have emerged. While their training and inference can be performed on large clusters, they remain way too big for ordinary machines like the ones most people use (only a few gigabytes of RAM). That is why reducing their size in terms of bytes has become a core field and a cutting-edge research topic. Often referred to as quantization, the main idea of this technique is to reduce the number of bits used to store each weight of a deep learning network, while minimizing the error this causes on the output of the model.

1.1 Context

Let $\mathcal{N}$ be a neural network made of n layers indexed from 1 to n (the very last layer). We want to optimize the number of bits b_i used to store the weights of each layer i, while naturally respecting a total budget.

Let $[a, c]$ be a segment of $\mathbb{R}$ in which the weights w of layer i are bounded, and let b be the number of bits used to quantize them. Since each bit can only take the value 0 or 1, b bits give us 2^b possible values. We define the step

$$\Delta = \frac{c-a}{2^b},$$

and we split the segment $[a, c]$ into 2^b subintervals of length Δ

$$I_k = [a + k\Delta,\ a + (k+1)\Delta], \qquad k = 0, \dots, 2^b - 1,$$

so that every weight belongs to one of them. We then map every weight $w \in I_k$ to its reconstruction level, defined as the middle of I_k:

$$q(w) = a + \left(k + \tfrac{1}{2}\right)\Delta.$$

We easily convince ourselves that the more bits we allocate, the smaller the step Δ, which also implies a smaller potential error, since the error is bounded by $\Delta/2$. The trade-off we have to balance between storage and precision leads us to a minimization problem: we want to minimize the error E, defined mathematically as a mean squared error (MSE), under a budget constraint. From there, we can prove that if we model the quantization error as a random variable uniformly distributed on $[-\Delta/2, \Delta/2]$, then the error is proportional to the variance of the weights times 2^{-2b}.

For a given layer i containing N_i weights $w_1, \dots, w_{N_i}$, we define the quantization error of each weight as $e_j = w_j - q(w_j)$, and we measure the error of the whole layer with the mean squared error (MSE):

$$E_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \big(w_j - q(w_j)\big)^2.$$

The optimization problem

Assuming that the errors of the different layers simply add up, the total error of the network is

$$E(b_1, \dots, b_n) = \sum_{i=1}^{n} E_i = \frac{\kappa^2}{3} \sum_{i=1}^{n} \sigma_i^2\, 2^{-2b_i}.$$

where $\kappa > 0$ is a fixed positive constant defining the quantization range. Since multiplying a function by a positive constant does not change its minimizers, we can drop the factor $\frac{\kappa^2}{3}$. Given a total budget of B bits, we finally consider the problem

$$\min_{b\in\mathbb{R}^n} J(b) = \sum_{i=1}^{n} \sigma_i^2\, 2^{-2b_i} \quad \text{subject to} \quad \sum_{i=1}^{n} b_i = B, \qquad b_i \geq 0,\ i = 1, \dots, n.$$

In this first approach, the b_i are allowed to take real values (continuous relaxation)

We define the equality constraint $G : \mathbb{R}^n \to \mathbb{R}$ and the inequality constraints $F_j : \mathbb{R}^n \to \mathbb{R}$, for $j = 1, \dots, n$, by

$$G(b) = \sum_{i=1}^{n} b_i - B, \qquad F_j(b) = -b_j.$$

The admissible set can then be written as

$$K = \left\{ b \in \mathbb{R}^n : \sum_{i=1}^{n} b_i = B,\ b_i \geq 0,\ i = 1, \dots, n \right\} = \left\{ b \in \mathbb{R}^n : G(b) = 0,\ F_j(b) \leq 0,\ j = 1, \dots, n \right\},$$

with one equality constraint and n inequality constraints. We assume $B \geq 0$.

K is non-empty. The vector $(B, 0, \dots, 0)$ satisfies $\sum_i b_i = B$ and $b_i \geq 0$ for all i, hence it belongs to K.

K is closed. The functions G and F_j are continuous (they are affine), so

$$K = G^{-1}(\{0\}) \cap \bigcap_{j=1}^{n} F_j^{-1}\big(]-\infty, 0]\big)$$

is a finite intersection of preimages of closed sets by continuous functions, hence a closed set.

K is bounded. For every $b \in K$, since all the b_i are non-negative,

$$\|b\|_1 = \sum_{i=1}^{n} |b_i| = \sum_{i=1}^{n} b_i = B,$$

so K is included in the closed ℓ^1 ball of radius B.

K is compact. K is a closed and bounded subset of $\mathbb{R}^n$, which is finite-dimensional, hence K is compact.

Existence of a minimizer. For every i, the function $b_i \mapsto \sigma_i^2\, 2^{-2b_i} = \sigma_i^2\, e^{-2b_i \ln 2}$ is continuous on $\mathbb{R}$, as the composition of an affine function and the exponential. Hence J, as a finite sum of continuous functions, is continuous on $\mathbb{R}^n$, and in particular on K.

Since J is continuous on the non-empty compact set K, the extreme value theorem (Weierstrass) ensures that J attains its minimum on K: there exists $b^* \in K$ such that

$$J(b^*) = \min_{b\in K} J(b).$$

2 Modelling

2.1 Convexity and uniqueness

We assume $\sigma_i^2 > 0$ for all i, i.e. no layer has all its weights equal.

J is strictly convex. For every i, let $\varphi_i(t) = \sigma_i^2\, 2^{-2t} = \sigma_i^2\, e^{-2t\ln 2}$. This function is twice differentiable on $\mathbb{R}$ and

$$\varphi_i''(t) = 4(\ln 2)^2\, \sigma_i^2\, e^{-2t\ln 2} > 0,$$

so φ_i is strictly convex. Since $J(b) = \sum_{i=1}^n \varphi_i(b_i)$, its Hessian matrix is diagonal:

$$\nabla^2 J(b) = \operatorname{diag}\left(\varphi_1''(b_1), \dots, \varphi_n''(b_n)\right),$$

with positive diagonal entries. It is therefore positive definite for every $b \in \mathbb{R}^n$, and J is strictly convex on $\mathbb{R}^n$.

K is convex. K is the intersection of the hyperplane $\{G = 0\}$ and of the half-spaces $\{F_j \le 0\}$, which are all convex sets. Hence K is convex.

Consequently, we conclude that J has a unique minimum as a function strictly convex on a convex.

2.2 Qualification of the constraints

We assume $B > 0$. The functions G and F_j are affine, hence differentiable on $\mathbb{R}^n$, with

$$G'(b) = (1, \dots, 1), \qquad F_j'(b) = -e_j, \qquad j = 1, \dots, n,$$

where $(e_1, \dots, e_n)$ denotes the canonical basis of $\mathbb{R}^n$. For $b \in K$, we denote by

$$I(b) = \{j \in \{1, \dots, n\} : F_j(b) = 0\} = \{j : b_j = 0\}$$

the set of active inequality constraints.

Linear independence of the equality constraints. There is a single equality constraint, and $G'(b) = (1, \dots, 1) \neq 0$. A family made of one non-zero vector is linearly independent.

Existence of a direction $\overline{w}$. We look for $\overline{w} \in [G'(b)]^\perp$, i.e. such that $\sum_{i=1}^n \overline{w}_i = 0$, satisfying for all $j \in I(b)$

$$\langle F_j'(b), \overline{w}\rangle = -\overline{w}_j < 0, \quad \text{i.e.} \quad \overline{w}_j > 0.$$

We choose the direction pointing towards the uniform allocation:

$$\overline{w} = \frac{B}{n}(1, \dots, 1) - b.$$

Then $\sum_{i=1}^n \overline{w}_i = B - \sum_{i=1}^n b_i = B - B = 0$, so $\overline{w} \in [G'(b)]^\perp$. Moreover, for every $j \in I(b)$, we have $b_j = 0$, hence $\overline{w}_j = \frac{B}{n} > 0$.

If $I(b) = \emptyset$, the second condition is automatically satisfied.

Therefore, the constraints are qualified at every point of K.

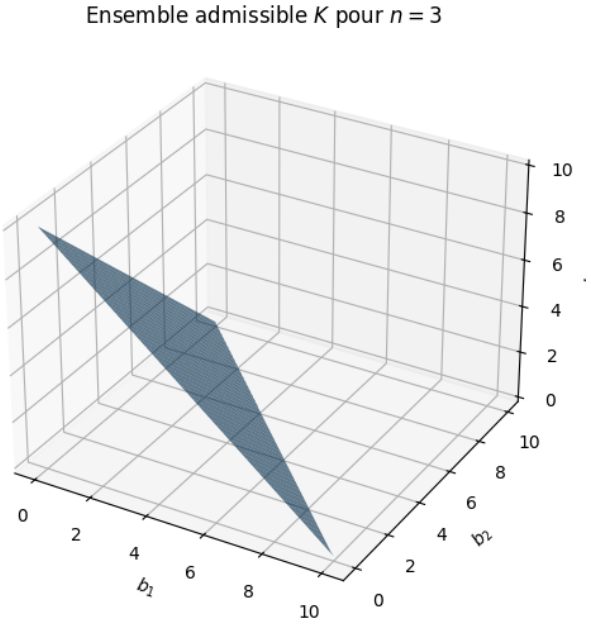


Figure 1. Admissible set $K = \{b \in \mathbb{R}^3 : b_1 + b_2 + b_3 = B,\ b_i \geq 0\}$ for $n = 3$ layers and $B = 10$. It is the triangle of the plane $\sum_i b_i = B$ cut out by the constraints $b_i \geq 0$, with vertices $(B, 0, 0)$, $(0, B, 0)$ and $(0, 0, B)$.

2.3 Karush–Kuhn–Tucker conditions

The general theorem. Let $u \in K$ be a local minimizer of J on K, such that J is differentiable at u and the constraints are qualified at u. Then there exist Lagrange multipliers $\mu = (\mu_1, \dots, \mu_N) \in \mathbb{R}^N$ and $\lambda = (\lambda_1, \dots, \lambda_M) \in \mathbb{R}_+^M$ such that

$$\begin{cases} J'(u) + \displaystyle\sum_{i=1}^{N} \mu_i\, G_i'(u) + \sum_{j=1}^{M} \lambda_j\, F_j'(u) = 0, \\ \lambda_j \geq 0, \quad \lambda_j F_j(u) = 0, \quad j = 1, \dots, M. \end{cases}$$

if the j-th constraint is not active ($F_j(u) < 0$), then $\lambda_j = 0$.

The Lagrangian. We introduce one multiplier $\mu \in \mathbb{R}$ for the budget constraint $G(b) = 0$, and n multipliers $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}_+^n$ for the constraints $F_j(b) = -b_j \leq 0$. The Lagrangian of the problem reads

$$\mathcal{L}(b, \mu, \lambda) = J(b) + \mu\, G(b) + \sum_{j=1}^{n} \lambda_j F_j(b) = \sum_{i=1}^{n} \sigma_i^2\, 2^{-2b_i} + \mu \left(\sum_{i=1}^{n} b_i - B \right) - \sum_{j=1}^{n} \lambda_j b_j.$$

Derivatives. For $i = 1, \dots, n$, since $2^{-2b_i} = e^{-2b_i \ln 2}$,

$$\frac{\partial J}{\partial b_i}(b) = -2 \ln 2\ \sigma_i^2\, 2^{-2b_i}, \qquad \frac{\partial G}{\partial b_i}(b) = 1, \qquad \frac{\partial F_j}{\partial b_i}(b) = -\delta_{ij},$$

where δ_{ij} is the Kronecker delta. Hence the i-th component of $\nabla_b \mathcal{L}$ is

$$\frac{\partial \mathcal{L}}{\partial b_i}(b, \mu, \lambda) = -2 \ln 2\ \sigma_i^2\, 2^{-2b_i} + \mu - \lambda_i.$$

The KKT system. The unique minimizer $b^* \in \mathbb{R}^n$ satisfies: there exist $\mu \in \mathbb{R}$ and $\lambda \in \mathbb{R}^n$ such that

$$\begin{cases} -2\ln 2\, \sigma_i^2\, 2^{-2b_i^*} + \mu - \lambda_i = 0, & i = 1, \dots, n \qquad \text{(stationarity)} \\ \displaystyle\sum_{i=1}^{n} b_i^* = B & \text{(budget constraint)} \\ b_i^* \geq 0, & i = 1, \dots, n \qquad \text{(primal feasibility)} \\ \lambda_i \geq 0, & i = 1, \dots, n \qquad \text{(dual feasibility)} \\ \lambda_i\, b_i^* = 0, & i = 1, \dots, n \qquad \text{(complementary slackness)} \end{cases}$$

2.4 Solving the KKT system

Step 1: case $b_i^* > 0$. By complementary slackness, $\lambda_i = 0$, and the stationarity equation becomes

$$\sigma_i^2\, 2^{-2b_i^*} = \frac{\mu}{2\ln 2}.$$

Step 2: the multiplier μ is positive. Since $B > 0$ and $\sum_i b_i^* = B$, at least one component $b_{i_0}^*$ is positive. Applying Step 1 to i_0 gives

$$\mu = 2\ln 2\, \sigma_{i_0}^2\, 2^{-2b_{i_0}^*} > 0.$$

We can therefore define the threshold

$$\theta = \frac{\mu}{2\ln 2} > 0.$$

Step 1 now reads: if $b_i^* > 0$, then $\sigma_i^2\, 2^{-2b_i^*} = \theta$, i.e.

$$b_i^* = \frac{1}{2}\log_2\left(\frac{\sigma_i^2}{\theta}\right),$$

which is positive if and only if $\sigma_i^2 > \theta$.

Step 3: case $b_i^* = 0$. The stationarity equation gives

$$\lambda_i = \mu - 2\ln 2\, \sigma_i^2 = 2\ln 2\, (\theta - \sigma_i^2),$$

and the condition $\lambda_i \geq 0$ imposes $\sigma_i^2 \leq \theta$.

Step 4: combining both cases. The two cases are mutually exclusive and exhaustive:

- if $\sigma_i^2 > \theta$, then $b_i^* = 0$ is impossible (it would require $\sigma_i^2 \leq \theta$), so $b_i^* > 0$;
- if $\sigma_i^2 \leq \theta$, then $b_i^* > 0$ is impossible (it would give $\sigma_i^2 = \theta\, 4^{b_i^*} > \theta$), so $b_i^* = 0$.

Hence, for every i,

$$\boxed{b_i^* = \max\left(0,\ \frac{1}{2}\log_2\frac{\sigma_i^2}{\theta}\right)}$$

and the multipliers are

$$\mu = 2\ln 2\ \theta, \qquad \lambda_i = 2\ln 2\ \max(0,\ \theta - \sigma_i^2).$$

Step 5: determining the threshold θ. The threshold is fixed by the budget constraint. Let

$$\Phi(\theta) = \sum_{i=1}^{n} \max\left(0,\ \frac{1}{2}\log_2 \frac{\sigma_i^2}{\theta}\right), \qquad \theta > 0.$$

The function Φ is continuous and non-increasing on $]0, +\infty[$. It is strictly decreasing on $]0, \sigma_{\max}^2[$, where $\sigma_{\max}^2 = \max_i \sigma_i^2$, and it vanishes on $[\sigma_{\max}^2, +\infty[$. Moreover, $\Phi(\theta) \to +\infty$ as $\theta \to 0^+$. Since $B > 0$, the intermediate value theorem ensures that there exists a unique $\theta^* \in]0, \sigma_{\max}^2[$ such that

$$\Phi(\theta^*) = B.$$

In practice, θ^* can be computed numerically.

Explicit formula. Let $A = \{i : \sigma_i^2 > \theta^*\}$ be the set of layers receiving bits, and $k = |A|$. The budget constraint gives

$$\sum_{i\in A} \frac{1}{2}\log_2 \frac{\sigma_i^2}{\theta^*} = B \quad \Longleftrightarrow \quad \theta^* = \left(\prod_{i\in A} \sigma_i^2\right)^{1/k} 2^{-2B/k}.$$

The threshold is the geometric mean of the variances of the active layers, scaled by $2^{-2B/k}$. In practice, we sort the variances in decreasing order and look for the largest k such that the corresponding θ^* satisfies $\sigma_i^2 > \theta^*$ for the k largest variances.

Particular case: all layers receive bits. If $A = \{1, \dots, n\}$, then

$$b_i^* = \frac{B}{n} + \frac{1}{2}\log_2 \frac{\sigma_i^2}{\overline{\sigma}^2}, \qquad \text{with } \overline{\sigma}^2 = \left(\prod_{j=1}^{n} \sigma_j^2\right)^{1/n}.$$

Each layer receives the average budget B/n, corrected according to its variance relative to the geometric mean of all variances.

3 Numerical resolution

3.1 Projected gradient method

The algorithm. Given an initial point $b^{(0)} \in K$ and a step $\rho > 0$, the projected gradient method builds the sequence

$$b^{(k+1)} = P_K\left(b^{(k)} - \rho\,\nabla J(b^{(k)})\right),$$

where P_K denotes the orthogonal projection onto K.

Existence of the projection. We showed that K is non-empty, closed and convex in the Hilbert space $\mathbb{R}^n$. By the projection theorem onto a closed convex set, for every $v \in \mathbb{R}^n$ there exists a unique

$P_K(v) \in K$ such that

$$\|v - P_K(v)\| = \min_{b \in K} \|v - b\|.$$

The algorithm is therefore well defined.

The projection is not trivial. Since K is the intersection of the hyperplane $\{\sum_i b_i = B\}$ and $\{b \geq 0\}$,

$$\min_{b \in \mathbb{R}^n} \frac{1}{2}\|b - v\|^2 \quad \text{subject to} \quad \sum_{i=1}^n b_i = B, \qquad b_i \geq 0,\ i = 1, \dots, n.$$

Solving the projection problem with KKT

Existence, uniqueness and qualification. The function $b \mapsto \frac{1}{2}\|b - v\|^2$ is continuous and strictly convex (its Hessian is the identity), and K is a non-empty convex compact set. Hence the problem has a unique solution. The constraints are the same as in our main problem, so they are qualified at every point of K and the objective is differentiable on $\mathbb{R}^n$. The KKT theorem applies.

The KKT system. We introduce a multiplier $\tau \in \mathbb{R}$ for the equality constraint and multipliers $\nu = (\nu_1, \dots, \nu_n) \in \mathbb{R}^n_+$ for the constraints $-b_i \leq 0$. The Lagrangian is

$$\mathcal{L}(b, \tau, \nu) = \frac{1}{2}\sum_{i=1}^n (b_i - v_i)^2 + \tau \left(\sum_{i=1}^n b_i - B \right) - \sum_{i=1}^n \nu_i b_i,$$

and the solution $p = P_K(v)$ satisfies

$$\begin{cases} p_i - v_i + \tau - \nu_i = 0, & i = 1, \dots, n, \\ \sum_{i=1}^n p_i = B, & \\ p_i \geq 0, \quad \nu_i \geq 0, \quad \nu_i\, p_i = 0, & i = 1, \dots, n. \end{cases}$$

Case $p_i > 0$. By complementary slackness, $\nu_i = 0$, so $p_i = v_i - \tau$, which requires $v_i > \tau$.

Case $p_i = 0$. The stationarity equation gives $\nu_i = \tau - v_i$, and the condition $\nu_i \geq 0$ imposes $v_i \leq \tau$.

Combining both cases. As in the main problem, the two cases are mutually exclusive and exhaustive: if $v_i > \tau$, then $p_i = 0$ would require $v_i \leq \tau$, which is impossible; if $v_i \leq \tau$, then $p_i > 0$ would give $p_i = v_i - \tau \leq 0$, which is impossible. Hence

$$\boxed{P_K(v)_i = \max(v_i - \tau,\, 0), \qquad i = 1, \dots, n,}$$

where τ is determined by the budget constraint

$$\Psi(\tau) = \sum_{i=1}^n \max(v_i - \tau,\, 0) = B.$$

Computing τ by sorting. Let $u_1 \geq u_2 \geq \cdots \geq u_n$ be the coordinates of v sorted in decreasing order. If exactly the k largest coordinates remain positive, the budget constraint gives

$$\sum_{j=1}^{k}(u_j - \tau) = B \quad \Longleftrightarrow \quad \tau = \frac{1}{k}\left(\sum_{j=1}^{k} u_j - B\right).$$

The correct k is the largest index such that $u_k - \frac{1}{k}\left(\sum_{j=1}^{k} u_j - B\right) > 0$. The projection can thus be computed exactly in $O(n \log n)$ operations, the cost being dominated by the sort.

Numerical results

We apply the projected gradient method to a three-layer example, with variances $\sigma^2 = (0.00255,\ 0.00781,\ 0.01562)$ and a budget $B = 12$, starting from the vertex $b^{(0)} = (0, 0, 12)$ with a step $\rho = 50$. Since all layers receive bits, the exact solution is given by the closed-form formula,

$$b^* = (3.295,\ 4.102,\ 4.602).$$

Qualitative behaviour. The path on K heads straight from the starting vertex towards b^*, and all iterates remain admissible, as guaranteed by the projection. The layers with the largest variances receive the most bits, as predicted by the KKT analysis.

Rate of convergence. After a fast initial phase, both $J(b^{(k)}) - J(b^*)$ and $\|b^{(k)} - b^*\|$ decrease along a straight line in logarithmic scale, which is the signature of the linear convergence predicted by the theorem. The measured error decreases from 9.08 to 1.52 after 500 iterations and to 0.51 after 1000 iterations, which corresponds to a contraction factor of about 0.9978 per iteration, i.e. roughly 1050 iterations to gain one digit of accuracy. After 1000 iterations, the method returns $b^{(1000)} = (3.087,\ 3.895,\ 5.018)$, still about half a bit away from b^*; reaching an accuracy of 10^{-10} requires about 7500 iterations.

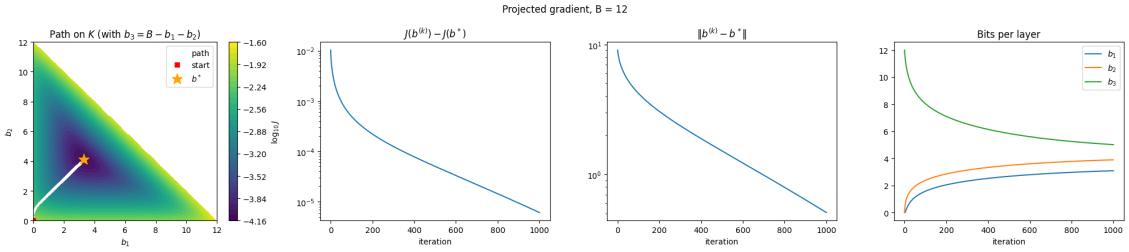


Figure 2. Projected gradient method for $\sigma^2 = (0.00255,\ 0.00781,\ 0.01562)$ and $B = 12$, with step $\rho = 50$, starting from $b^{(0)} = (0, 0, 12)$ (1000 iterations). From left to right: path of the iterates on K over the level sets of $\log_{10} J$, gap $J(b^{(k)}) - J(b^*)$ and error $\|b^{(k)} - b^*\|$ (logarithmic vertical axis), and number of bits allocated to each layer along the iterations.

3.2 Uzawa's algorithm

Principle. Instead of enforcing the constraints on b at each iteration, Uzawa's algorithm looks for a saddle point of the Lagrangian

$$\mathcal{L}(b, \mu, \lambda) = J(b) + \mu\, G(b) + \lambda \cdot F(b), \qquad \mu \in \mathbb{R},\ \lambda \in \mathbb{R}^n_+,$$

by alternating a minimization in the primal variable b and a projected gradient ascent in the multipliers (μ, λ).

Algorithm. Given $\mu_0 > 0$, $\lambda_0 = 0$ and a step $\rho > 0$, repeat for $k \geq 0$:

1. **Primal step.** Minimize the Lagrangian without constraint:

$$b^{(k)} = \arg\min_{b \in \mathbb{R}^n} \mathcal{L}(b, \mu_k, \lambda_k).$$

   Since $\mathcal{L}$ is separable in b, this minimization is explicit:

$$b_i^{(k)} = \frac{1}{2} \log_2 \left(\frac{2 \ln 2 \, \sigma_i^2}{\mu_k - \lambda_{k,i}} \right), \qquad i = 1, \dots, n.$$

2. **Dual step.** Update the multipliers with a gradient ascent, projected onto $\mathbb{R}_+$ for the inequality constraints:

$$\mu_{k+1} = \mu_k + \rho \, G(b^{(k)}), \qquad \lambda_{k+1} = \max\left(0, \ \lambda_k + \rho \, F(b^{(k)})\right).$$

The multiplier μ can be interpreted as the price of a bit: if the layers request more bits than the budget allows ($G(b^{(k)}) > 0$), the price increases, otherwise it decreases. We initialize it with the marginal gain of a bit at the uniform allocation,

$$\mu_0 = \frac{1}{n} \sum_{i=1}^{n} -\frac{\partial J}{\partial b_i} \left(\frac{B}{n}, \dots, \frac{B}{n} \right),$$

which is a natural estimate of $\mu^* = -\frac{\partial J}{\partial b_i}(b^*)$ that does not use the closed-form solution.

Numerical results

We apply Uzawa's algorithm to the same example as the projected gradient method with

$$\sigma^2 = (0.00255, \ 0.00781, \ 0.01562)$$

with a step $\rho = 10^{-5}$. The algorithm stops after only 26 iterations, with $\|b^{(26)} - b^*\| \approx 4 \times 10^{-11}$, and the gradient of the Lagrangian vanishes at machine precision. The final multipliers are $\lambda = 0$, since no constraint $b_i \geq 0$ is active, and $\mu \approx 3.67 \times 10^{-5}$, which coincides with the value $2 \ln 2 \, \theta^*$ predicted by the KKT analysis.

Rate of convergence. Both $J(b^{(k)}) - J(b^*)$ and $\|b^{(k)} - b^*\|$ decrease along a straight line in logarithmic scale, so the convergence is again linear, but with a contraction factor of about 0.41 per iteration, i.e. roughly 2.6 iterations per digit of accuracy. For comparison, the projected gradient method had a contraction factor of 0.9978 and required about 7500 iterations to reach a similar accuracy.

Interpretation. This difference has a simple explanation. The primal step is solved exactly, so the only unknown left is the scalar multiplier μ (as long as $\lambda = 0$). Near the optimum, the dual step

reads $\mu_{k+1} = \mu_k + \rho\,(\sum_i b_i(\mu_k) - B)$, and since

$$\frac{d}{d\mu}\sum_{i=1}^{n} b_i(\mu) = -\frac{n}{2\ln 2\;\mu},$$

the local contraction factor is

$$1 - \rho\,\frac{n}{2\ln 2\;\mu^*} \approx 0.41,$$

which matches the measured value exactly. Uzawa's algorithm thus solves a one-dimensional problem, whatever the number of layers, instead of moving in $\mathbb{R}^n$ along the very flat directions of J that slow down the projected gradient method.

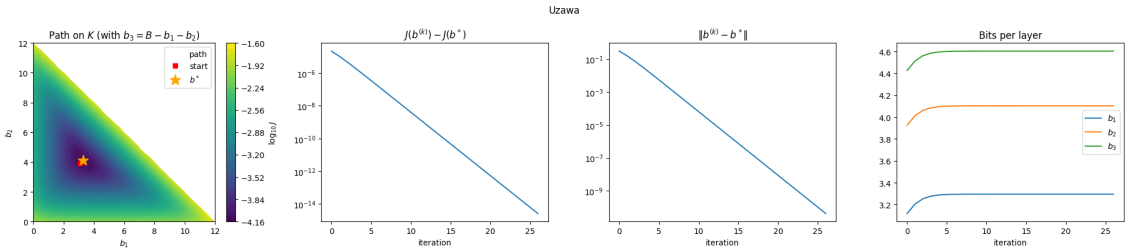


Figure 3. Uzawa's algorithm for $\sigma^2 = (0.00255,\ 0.00781,\ 0.01562)$ and $B = 12$, with step $\rho = 10^{-5}$, $\lambda_0 = 0$ and μ_0 given by the marginal gain at the uniform allocation (26 iterations). From left to right: iterates in the plane (b_1, b_2) over the level sets of $\log_{10} J$, gap $J(b^{(k)}) - J(b^*)$ and error $\|b^{(k)} - b^*\|$ (logarithmic vertical axis), and number of bits allocated to each layer along the iterations.

4 Conclusion

	Projected gradient	**Uzawa**
Working space	primal, $b \in \mathbb{R}^n$	dual, $(\mu, \lambda) \in \mathbb{R} \times \mathbb{R}^n_+$
Step ρ	50	10^{-5}
Initialization	$b^{(0)} = (0, 0, 12)$	μ_0 = marginal gain at $\frac{B}{n}$, $\lambda_0 = 0$
Cost per iteration	$O(n \log n)$ (projection by sorting)	$O(n)$ (explicit minimization)
Admissible iterates	yes, $b^{(k)} \in K$	no, only at the limit
Multipliers provided	no	yes, $\mu \approx 3.67 \times 10^{-5}$, $\lambda = 0$
Contraction factor	0.9978	0.41
Iterations per digit	≈ 1050	≈ 2.6
Iterations to reach 10^{-10}	≈ 7500	26

Table 1. Comparison of the projected gradient method and Uzawa's algorithm on the example $\sigma^2 = (0.00255,\ 0.00781,\ 0.01562)$, $B = 12$.

Table 1 summarizes the behaviour of both methods on the same example: Uzawa's algorithm converges about 300 times faster, since it only solves a one-dimensional dual problem, but its iterates are not admissible until convergence, whereas the projected gradient method always returns a valid bit allocation.

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[2] R. Marques Araujo, *optimal-bit-allocation*: source code of the project, GitHub repository, 2026. https://github.com/Debian-code/optimal-bit-allocation